Artificial Intelligence

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Overview

Intelligence

Artificial Intelligence

Types of A.I.

Components of A.I.

Applications of A.I.

Intelligence

Intelligence

"The ability to acquire and apply knowledge."

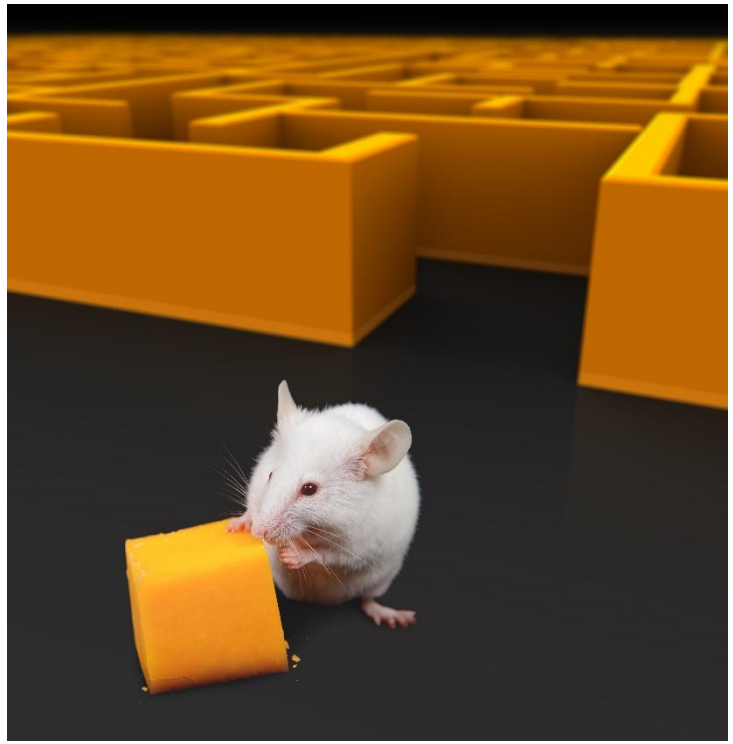
"The ability to learn or understand."

"Goal-directed adaptive behavior."

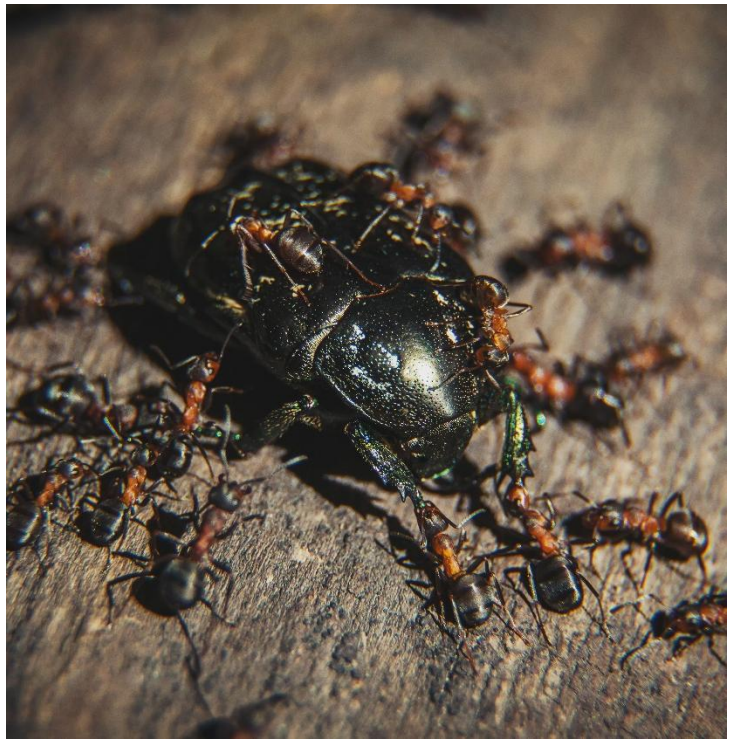
"The ability to deal with cognitive complexity."

Examples of Intelligence

Human intelligence



Animal intelligence



Collective intelligence

Examples of Intelligence

	Agent
	Human
	Mouse
	Ants

Examples of Intelligence

	Agent	Environment
	Human	Exam
	Mouse	Maze
	Ants	Colony

Examples of Intelligence

	Agent	Environment	Goal
	Human	Exam	Pass
	Mouse	Maze	Cheese
	Ants	Colony	Protect

Examples of Intelligence

	Agent	Environment	Goal	Perception
	Human	Exam	Pass	Read
	Mouse	Maze	Cheese	See
	Ants	Colony	Protect	Smell

Examples of Intelligence

	Agent	Environment	Goal	Perception	Action
	Human	Exam	Pass	Read	Solve
	Mouse	Maze	Cheese	See	Navigate
	Ants	Colony	Protect	Smell	Attack

Examples of Intelligence

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Intelligence

The ability of an agent
to perceive an environment
and to choose actions
that increase its chances
of achieving a goal

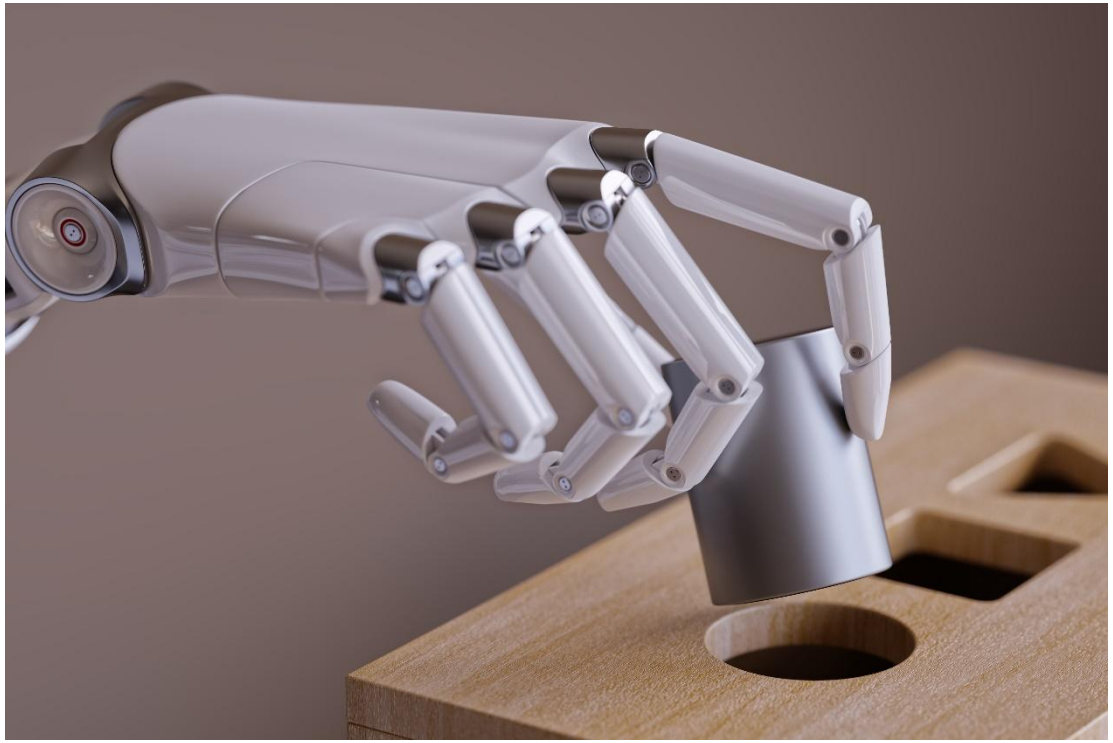
Intelligence

The ability of an agent
to perceive an environment
and to choose actions
that increase its chances
of achieving a goal
by learning, knowledge,
reasoning, planning, etc.

Artificial Intelligence

Types of Intelligence

Natural intelligence



Artificial intelligence

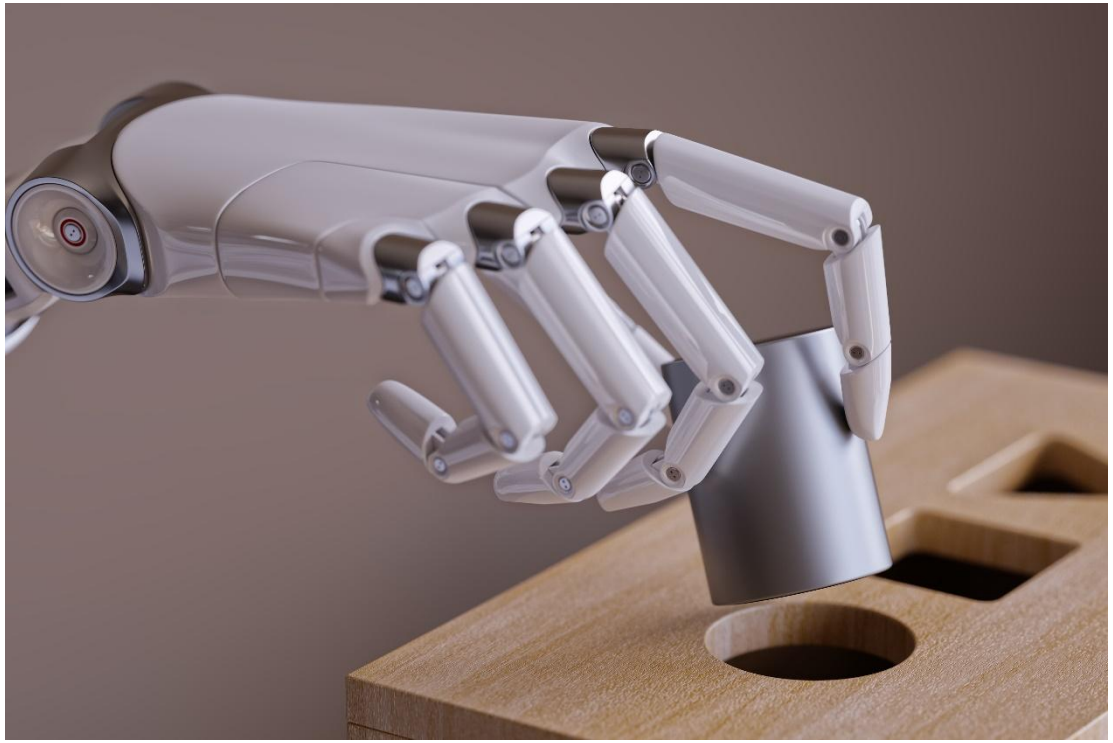
Types of Intelligence

Natural intelligence

Artificial intelligence

Types of Intelligence

Natural intelligence



Artificial intelligence

Artificial Intelligence

The ability of a machine
to replicate natural intelligence

Artificial Intelligence

The ability of a machine
to perceive an environment
and to choose actions
that maximize
the expected likelihood
of achieving a goal

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Artificial Intelligence

Anything a human can do
but a machine cannot yet do

Types of A.I.

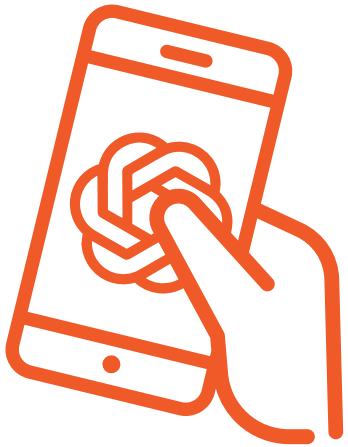
Types of A.I.

Artificial narrow intelligence

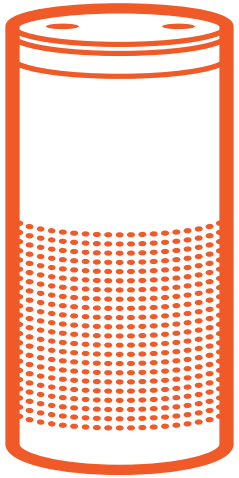
Artificial general intelligence

Artificial super intelligence

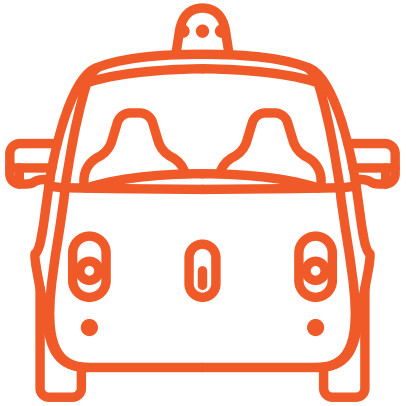
Artificial Narrow Intelligence



OpenAI
ChatGPT

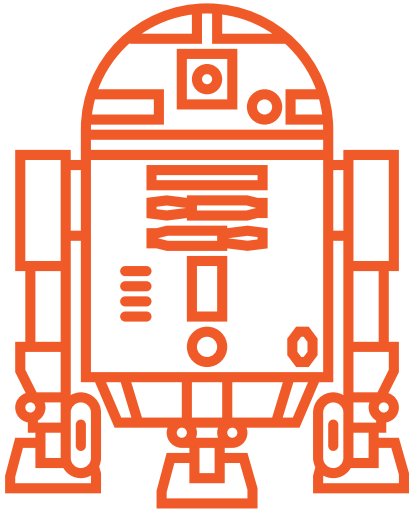


Amazon
Alexa

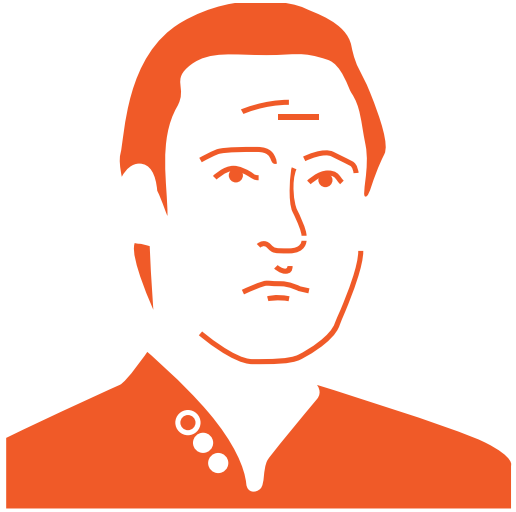


Google
Waymo

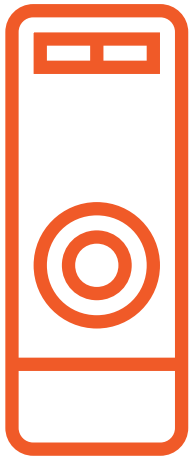
Artificial General Intelligence



R2D2



Data



HAL 9000

Artificial General Intelligence

AI that can do 50% of all economically valuable work as well or better than 50% of all humans.

Sparks of Artificial General Intelligence: Early experiments with GPT-4

Sébastien Bubeck Varun Chandrasekaran Ronen Eldan Johannes Gehrke
Eric Horvitz Ece Kamar Peter Lee Yin Tat Lee Yuanzhi Li Scott Lundberg
Harsha Nori Hamid Palangi Marco Tulio Ribeiro Yi Zhang

Microsoft Research

Abstract

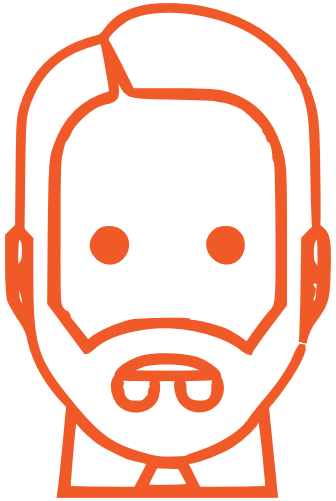
Artificial intelligence (AI) researchers have been developing and refining large language models (LLMs) that exhibit remarkable capabilities across a variety of domains and tasks, challenging our understanding of learning and cognition. The latest model developed by OpenAI, GPT-4 [Ope23], was trained using an unprecedented scale of compute and data. In this paper, we report on our investigation of an early version of GPT-4, when it was still in active development by OpenAI. We contend that (this early version of) GPT-4 is part of a new cohort of LLMs (along with ChatGPT and Google's PaLM for example) that exhibit more general intelligence than previous AI models. We discuss the rising capabilities and implications of these models. We demonstrate that, beyond its mastery of language, GPT-4 can solve novel and difficult tasks that span mathematics, coding, vision, medicine, law, psychology and more, without needing any special prompting. Moreover, in all of these tasks, GPT-4's performance is strikingly close to human-level performance, and often vastly surpasses prior models such as ChatGPT. Given the breadth and depth of GPT-4's capabilities, we believe that it could reasonably be viewed as an early (yet still incomplete) version of an artificial general intelligence (AGI) system. In our exploration of GPT-4, we put special emphasis on discovering its limitations, and we discuss the challenges ahead for advancing towards deeper and more comprehensive versions of AGI, including the possible need for pursuing a new paradigm that moves beyond next-word prediction. We conclude with reflections on societal influences of the recent technological leap and future research directions.

Source: Microsoft Research (2023)

Artificial Super Intelligence

Skynet

Ultron



The Architect

Types of A.I.

Artificial narrow intelligence	Artificial general intelligence	Artificial super intelligence

Components of A.I.

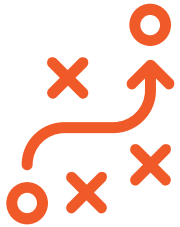
Components of Artificial Intelligence

Perception

Learning

Knowledge

Reasoning



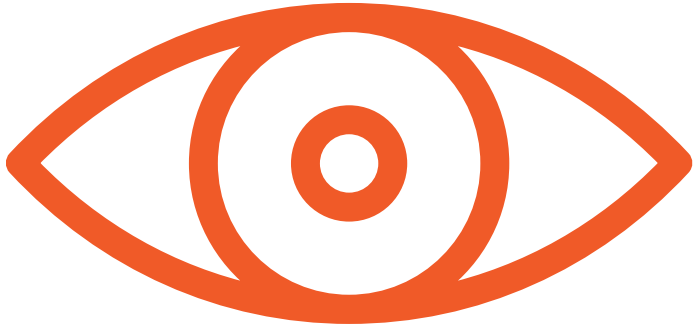
Planning

Perception

Deduce state from sensors

Recognize patterns

Detect features



Learning

Extract knowledge

Data in, action out

Maps state to action

Knowledge

Represent learning

Collection of information

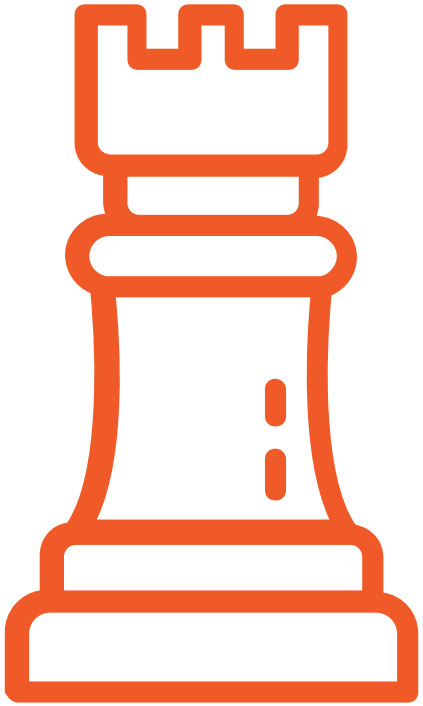
Various representations

Reasoning

Infer conclusions

Deduction

Induction

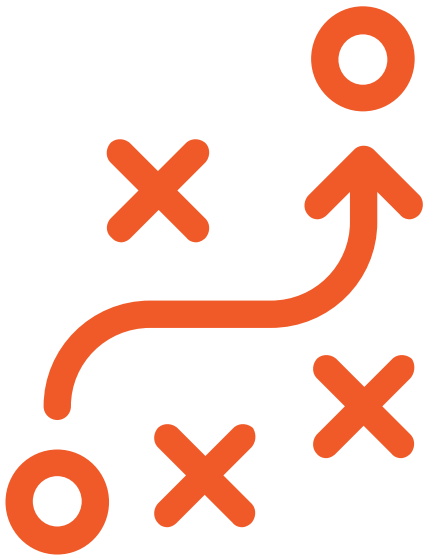


Planning

Set and achieve goals

Visualize future states

Predict actions



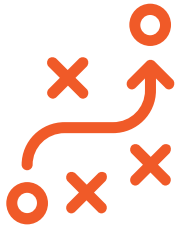
Components of Artificial Intelligence

Perception

Learning

Knowledge

Reasoning



Planning

Applications of A.I.

Examples of A.I.

Customer Service

FAQ generators

Chat bots

Voice assistants

Finance

Trading algorithms

Fraud detection

Portfolio management

Healthcare

Diagnostic tools

Treatment recommendation

Prescription verification

Manufacturing

Product design

Industrial robots

Defect detection

Marketing

Advertisement optimization

Sentiment analysis

Product recommendation

Transportation

Warehouse robots

Route optimization

Delivery drones

This is just the tip of the iceberg!

Summary

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Intelligence

Artificial Intelligence

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